

TEXAS INSTRUMENTS
INCORPORATED
AUSTIN, TEXAS

DOCUMENT NUMBER REVISION
064801-0002 *C

INTIMER - INTERVAL TIMER PDT - 900

SHEET 0002

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0001 * TITLE=INTIMER - INTERVAL TIMER PDT - 900
0002 * AUTHOR =
0003 *
0004 * REVISION=07/30/75 - BOB LEDBETTER
0005 * * REWRITE TO REMOVE UNNEEDED CODE GENERATED BY METL
0006 * * CHANGED TO USE TTY/ETA
0007 *
0008 * REVISION=03/27/74 - DON ROBERTSON
0009 * * MODIFIED TO RUN ON EITHER TTY OR ETA INTERFACE
0010 *
0011 * COMPUTER=980,SAPG
0012 IDT INTIMER
0013 HED INTIMER - INTERVAL TIMER PDT - 900
0000 0014 A EQU 0
0001 0015 E EQU 1
0002 0016 X EQU 2
0003 0017 M EQU 3
0004 0018 S EQU 4
0005 0019 B EQU 6
0001 0020 BR EQU 1
0008 0021 ST EQU 8
808A 0022 CRLF EQU >AD8A
0023 REF WRITER,RECONS,CONVRS,WAITS
0000 8000 0024 START DLD S+1 SET UP PARITY ERROR TRAP
0001 CF00 0025 IDL 0 LOCATION
0002 78FF 0026 BRU S-1
0003 A702 0027 DST #2
0004 D880 0028 LSB 0 ENABLE PARITY INTERRUPTS
P 0005 0006 0029 DATA S+1,STARTC,>10
P 0006 0047
0007 0010
P 0008 0000 0030 STARTS DATA START
0000 0031 LOCR EQU S
P 0009 0005 0032 DATA RDS1
P 000A 00AF 0033 DATA RDS2
P 000B 00C5 0034 DATA RDS3
P 000C 00D0 0035 DATA RDS4
P 000D 00E4 0036 DATA RDS5
P 000E 0101 0037 DATA RDS6
P 000F 0119 0038 DATA RDS7
P 0010 0124 0039 DATA RDS8
P 0011 013F 0040 DATA RDS9
P 0012 014C 0041 DATA RDS10
P 0013 015F 0042 DATA RDS11
P 0014 018A 0043 DATA RDS12
P 0015 01A5 0044 DATA RDS13
P 0016 01B3 0045 DATA RDS14
P 0017 01ED 0046 DATA RDS15
P 0018 01FA 0047 DATA RDS16
P 0019 020C 0048 DATA RDS17
P 001A 0242 0049 DATA RDS18
P 001B 0250 0050 DATA RDS19
P 001C 0270 0051 DATA RDS20

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Change 1

TEXAS INSTRUMENTS
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DOCUMENT NUMBER REVISION
068801-9902 *C

INTER - INTERVAL TIMER PDT - 990

SHEET 0003

P	0010	0287	0052		DATA	RDS21
P	001F	028J	0053		DATA	RDS22
P	001F	02C0	0054		DATA	RDS23
		0020	0055	LOCW	EQU	S
P	0020	0093	0056		DATA	WDS1
P	0021	00AD	0057		DATA	WDS2
P	0022	00E2	0058		DATA	WDS3
P	0023	00FA	0059		DATA	WDS4
P	0024	00FF	0060		DATA	WDS5
P	0025	013C	0061		DATA	WDS6
P	0026	015C	0062		DATA	WDS7
P	0027	017F	0063		DATA	WDS8
P	0028	018F	0064		DATA	WDS9
P	0029	019A	0065		DATA	WDS10
P	002A	01AC	0066		DATA	WDS11
P	002B	01C2	0067		DATA	WDS12
P	002C	01CA	0068		DATA	WDS13
P	002D	01D5	0069		DATA	WDS14
P	002F	01DF	0070		DATA	WDS15
P	002F	0234	0071		DATA	WDS16
P	0030	023A	0072		DATA	WDS17
P	0031	0249	0073		DATA	WDS18
P	0032	0262	0074		DATA	WDS19
P	0033	026B	0075		DATA	WDS20
P	0034	0271	0076		DATA	WDS21
P	0035	0280	0077		DATA	WDS22
P	0036	0296	0078		DATA	WDS23
P	0037	029C	0079		DATA	WDS24
P	0038	02A5	0080		DATA	WDS25
P	0039	02A9	0081		DATA	WDS26
P	003A	02B8	0082		DATA	WDS27
P	003B	02CF	0083		DATA	WDS28
P	003C	022B	0084		DATA	WDS29
		003D	0085	LOCS	EQU	S
P	003D	00F1	0086	TEST2S	DATA	TEST2
P	003E	017A	0087	TEST3S	DATA	TEST3
P	003F	01D0	0088	TEST4S	DATA	TEST4
P	0040	0226	0089	TEST5S	DATA	TEST5
		0090	0090	WRITE	DATA	WRITE
X	0041	0000				
		0001	0091	RECON	DATA	RECON
X	0042	0000				
		0002	0092	CONWR	DATA	CONWR
X	0043	0000				
		0003	0093	WAIT	DATA	WAIT
X	0044	0000				
P	0045		0094	GROUP	RSS	1
P	0046		0095	REGADD	RSS	1
	0047	74F9	0096	STARTC	RRI	*WRITE
P	0048	020P	0097		DATA	DOS
	0049	74F7	0098	STARTA	RRI	*WRITE
P	004A	02F5	0099		DATA	D13
	004B	0F01	0100		LDI	#1

PRINT HEADER
REQUEST GROUP

INTMER - INTERVAL TIMER PDT - 9AP

SHEET 0004

0040	74F5	0101		RRL	*RECON	READ GROUP
0041	6700	0102		CPL	#0	GROUP >#0?
0045	C0C0	0103		SLE	A	
0046	78F0	0104		RRU	STARTA	NO - REQUEST GROUP AGAIN
0050	6703	0105		CPL	#3	GROUP <#3?
0051	C080	0106		SGF	A	
0052	78F6	0107		RRU	STARTA	NO - REQUEST AGAIN
0053	87E1	0108		STA	GROUP	
0054	74EC	0109	STARTA	RRL	*WRITE	REQUEST REG ADDRESS
P 0055	0315	0110		DATA	D35	
0056	0F01	0111		LDE	#1	
0057	74EA	0112		RRL	*RECON	READ REG ADDRESS
0058	6700	0113		CPL	#0	REG ADDRESS >#0?
0059	C0C0	0114		SLE	A	
005A	78F0	0115		RRU	STARTA	NO - REQUEST ADDRESS AGAIN
005B	675F	0116		CPL	#5F	REG ADDRESS <#5F?
005C	C080	0117		SGF	A	
005D	78F6	0118		RRU	STARTA	NO - REQUEST ADDRESS AGAIN
005E	80E7	0119		STA	REGADD	
005F	00E5	0120		LDA	GROUP	BUILD GROUP AND REG ADDRESS
0060	C0C0	0121		LLA	0	FOR RDS - WDS COMMANDS
0061	32E4	0122		TOR	REGADD	
0062	17E0	0123		LDX	#LOCW-LOCW	# OF RDS INSTRUCTIONS TO MOD
0063	3100	0124		*IDR	#D000	
0064	0000					
0065	0A8A	0125	LOOP1	LDE	LOCW,X	LOC OF RDS INSTR
0066	C516	0126		RMO	E,B	PUT LOC IN BASE REG
0067	8100	0127		STA	0,BR	STORE IT
0068	40FC	0128		RIX	LOOP1	
0069	17E3	0129		LDX	#LOCW-LOCW	# OF WDS INSTRUCTIONS TO MOD
006A	3720	0130		TOR	#20	
006B	0A01	0131	LOOP2	LDE	LOCW,X	LOC OF WDS INSTR
006C	C516	0132		RMO	E,B	PUT LOC IN BASE REG
006D	8100	0133		STA	0,BR	STORE IT
006E	40FC	0134		RIX	LOOP2	
006F	0098	0135		LDA	STARTA	SET UP BASE REG FOR USE
0070	C506	0136		RMO	A,B	IN PROGRAM
0071	1800	0137		OLDM	#110	STATUS FOR I/O INTERRUPTS
0072	0110					
0073	0710	0138		LDA	#10	
0074	C504	0139		RMO	A,S	STATUS FOR NO I/O INTERRUPTS
0075	74C0	0140	TSTSEL	RRL	*WRITE	REQUEST TEST NUMBER
P 0076	0330	0141		DATA	D35	
0077	0F01	0142		LDE	#1	
0078	74C0	0143		RRL	*RECON	READ TEST NUMBER
0079	7541	0144		RRL	*WRITE,BR	
P 007A	041A	0145		DATA	D15A5	
007B	6701	0146		CPL	#1	TEST 1?
007C	C0A0	0147		SNF	A	YES
007D	780F	0148		RRU	TEST1	NO - TEST 2?
007E	6702	0149		CPL	#2	
007F	C0A0	0150		SNF	A	YES
0080	7C0C	0151		RRU	*TEST28	

TEXAS INSTRUMENTS
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DOCUMENT NUMBER REVISION
966001-9902 *C

INTMR - INTERVAL TIMER PDT - 990

SHEET 0005

0001	6703	0152	CPL	#3	NO - TEST 3?
0002	CDAP	0153	SNF	A	
0003	7CBA	0154	PRU	*TEST38	YES
0004	6704	0155	CPL	#4	NO - TEST 4?
0005	CDAP	0156	SNF	A	
0006	7CBA	0157	PRU	*TEST48	YES
0007	6705	0158	CPL	#5	NO - TEST 5?
0008	CDAP	0159	SNF	A	
0009	7CBA	0160	PRU	*TEST58	YES
000A	74BA	0161	PRU	*WRITE	NO - ERROR
P 000B	0371	0162	DATA	D43	
000C	78ER	0163	PRU	TSTSEI	

TEXAS INSTRUMENTS
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DOCUMENT NUMBER REVISTON
986801-0002 *C

INTMER - INTERVAL TIMER PNT - 9A0

SHEET 0006

		0164	PEJ	
		0165	*****	
		0166	* TEST 1 - START/STOP TEST	
		0167	*****	
		0168	TEST1 SSN 8	IF SS#1 IS ON DO NOT PRINT
		0169	RRU TEST1A	
		0170	RRL *WRITE	WRITE TEST TITLE
		0171	DATA D53	
		0172	TEST1A RMD S,ST	DISABLE I/O INTERRUPTS
		0173	LDE #0	
		0174	WDS1 WDS 0	OUTPUT STOP COMMAND
		0175	DATA 1	
		0176	RDS1 RDS 0	READ STATUS
		0177	DATA 0	
		0178	*AND =>F000	LOOK AT 4 MSB'S
		0179	*CPL #0	
		0180	SNE A	
		0181	RRU TEST1B	YES
		0182	TEST1F SSN 8	IF SS#1 IS ON DO NOT PRINT
		0183	RRU TEST1C	
		0184	RRL *WRITE	PRINT ERROR MESSAGE
		0185	DATA D53	
		0186	LDE #1	
		0187	RRL *CONWR	
		0188	TEST1C SSE 4	IF SS#2 IS ON REPEAT TEST
		0189	RRU TSTSEL	OTHERWISE RETURN TO TEST SEL
		0190	RRU TEST1A	
		0191	TEST1B OLD S+1	SET UP I/O TRAP ADDRESS
		0192	*SSB T1INT	
		0193	DST #6	
		0194	RMD H,ST	ENABLE I/O INTERRUPTS
		0195	*LDE =>400F	START TIMER/CNT=15
		0196	WDS2 WDS 0	
		0197	DATA 1	
		0198	RDS2 RDS 0	READ TIMER
		0199	DATA 0	
		0200	*CPL =>100F	CORRECT DATA?
		0201	SNE A	
		0202	RRU TEST1D	YES
		0203	RMD S,ST	DISABLE I/O INTERRUPTS
		0204	SSN 8	IF SS#1 IS ON DO NOT PRINT
		0205	RRU TEST1E	
		0206	RRL *WRITE	PRINT ERROR MESSAGE
		0207	DATA D73	
		0208	LDE #1	
		0209	RRL *CONWR	
		0210	TEST1F SSE 4	IF SS#2 IS ON REPEAT TEST
		0211	RRU TSTSEL	OTHERWISE RETURN TO TEST SEL

TEXAS INSTRUMENTS
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DOCUMENT NUMBER REVISION

966801-9902

*C

INTHER - INTERVAL TIMER PDT - 990

SHEET 0007

000E	7A02	0212	RRU	TEST1A	
000F	0000	0213	TEST1D	ALDA	#200
00C0	00CA				
00C1	7A02	0214	RRL	*WAIT	
00C2	C540	0215	PMO	S,ST	
00C3	CC0A	0216	SSN	8	
00C4	7A06	0217	RRU	TEST1F	
00C5	DA00	0218	RDS3	RDS	0
00C6	0000	0219		DATA	0
00C7	7541	0220	RRL	*WRITE,RR	
P 00CA	03A9	0221		DATA	D03
00C9	0F01	0222		LOE	#1
00CA	7543	0223	RRL	*CONWR,RR	
00CB	CC14	0224	TEST1F	SSE	4
00CC	7A0A	0225	RRU	TSTSEL	
00CD	7A03	0226	RRU	TEST1A	
00CE	0000	0227	T1INT	DATA	0,0
00CF	0000				
00D0	DA00	0228	RDS4	RDS	0
00D1	0000	0229		DATA	0
00D2	3A00	0230		*AND	#>F000
00D3	F000				
00D4	6000	0231		*CPL	#>0000
00D5	9000				
00D6	CD0A	0232	SNE	A	
00D7	7A09	0233	RRU	TEST1G	
00D8	CC0A	0234	SSN	8	
00D9	7A04	0235	RRU	TEST1H	
00DA	7541	0236	RRL	*WRITE,RR	
P 00DB	03B7	0237		DATA	D03
00DC	0F01	0238		LOE	#1
00DD	7543	0239	RRL	*CONWR,RR	
00DE	CC14	0240	TEST1H	SSE	4
00DF	7A95	0241	BRU	TSTSEL	
00E0	7A00	0242	RRU	TEST1A	
00E1	0F00	0243	TEST1G	LOE	#0
00E2	DA20	0244	WDS3	WDS	0
00E3	0F01	0245		DATA	1
00E4	DA00	0246	RDS5	RDS	0
00E5	0000	0247		DATA	0
00E6	3A00	0248		*AND	#>F000
00E7	F000				
00E8	6700	0249	CPL	#0	
00E9	CD20	0250	SEC	A	
00EA	7A02	0251	RRU	TEST1Y	
00EB	7541	0252	RRL	*WRITE,RR	
P 00EC	03CC	0253		DATA	D103
00ED	CC04	0254	SSN	4	
00EF	7A02	0255	RRU	TEST1A	
00EF	CC02	0256	SSN	2	
00F0	7975	0257	RRU	TSTSEL,RR	

WAIT 200MS FOR INTERRUPT
DISABLE I/O INTERRUPTS
IF SS#1 IS ON DO NOT PRINT

READ TIMER

PRINT ERROR MESSAGE

IF SS#2 IS ON REPEAT TEST
OTHERWISE RETURN TO TEST SEL

TEST 1 INTERRUPT ROUTINE

READ TIMER

LOOK AT STATUS ONLY

CORRECT?

YES
IF SS#1 IS ON DO NOT PRINT

PRINT ERROR MESSAGE

IF SS#2 IS ON REPEAT TEST
OTHERWISE RETURN TO TEST SEL

SEND STOP COMMAND

READ STATUS

LOOK AT STATUS ONLY

CORRECT?

NO
PRINT TEST OK

IF SS#2 IS ON REPEAT TEST

IF SS#3 ON RETURN TO TEST SEL

TEXAS INSTRUMENTS
INCORPORATED
AUSTIN, TEXAS

DOCUMENT NUMBER REVISION
966801-0002 *C

INTHER - INTERVAL TIMER POT - 900

SHEET 0000

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0258 PEJ
0259 *****
0260 * TEST 2 - LOAD AND START TEST
0261 *****
00F1 CC90 0262 TEST2 SSN 8 IF SS#1 IS ON DO NOT PRINT
00F2 7A02 0263 BRU TEST2A
00F3 7541 0264 BRU *WRITE,RR PRINT TEST TITLE
P 00F4 0305 0265 DATA 0119
00F5 0F00 0266 TEST2A LDE #0
00F6 0A20 0267 WDS4 WDS 0 OUTPUT STOP COMMAND
00F7 0001 0268 DATA 1
00F8 0000 0269 DLD S+1 LOAD I/O TRAP ADDRESS
00F9 0AC0 0270 *SSB T2INTA
P 00FA 0120
00FB A706 0271 DST #6
00FC C53A 0272 RMO M,ST ENABLE I/O INTERRUPTS
00FD 0000 0273 *LDE #400F
00FE 400F
00FF 0A20 0274 WDS5 WDS 0 START TIMER/CNT=15
0100 0001 0275 DATA 1
0101 0A00 0276 RDS4 RDS 0 READ TIMER
0102 0000 0277 DATA 0
0103 3A00 0278 *AND #F000 LOOK AT STATUS ONLY
0104 F000
0105 0000 0279 *CPL #1000 CORRECT?
0106 1000
0107 C0A0 0280 SNE A
0108 7A0A 0281 BRU TEST2B YES
0109 C54A 0282 TEST2J RMO S,ST DISABLE I/O INTERRUPTS
010A CC90 0283 SSN 8 IF SS#1 IS ON DO NOT PRINT
010B 7A04 0284 BRU TEST2C
010C 7541 0285 BRU *WRITE,RR PRINT ERROR MESSAGE
P 010D 030A 0286 DATA 073
010E 0F01 0287 LDE #1
010F 7543 0288 BRU *CONWR,RR
0110 CC14 0289 TEST2C SSE 4 IF SS#2 IS ON REPEAT TEST
0111 7075 0290 BRU TSTSEL,RR OTHERWISE RETURN TO TEST SEL
0112 7AE2 0291 BRU TEST2A
0113 0000 0292 TEST2B *LDA #200
0114 00CA
0115 7544 0293 BRU *WAIT,RR WAIT 200MS FOR I/O INTERRUPT
0116 C54A 0294 TEST2E RMO S,ST DISABLE I/O INTERRUPTS
0117 CC90 0295 SSN 8 IF SS#1 IS ON DO NOT PRINT
0118 7A0A 0296 BRU TEST2F
0119 0A00 0297 RDS7 RDS 0 READ TIMER
011A 0000 0298 DATA 0
011B 7541 0299 BRU *WRITE,RR PRINT ERROR MESSAGE
P 011C 03A0 0300 DATA 0B3
011D 0F01 0301 LDE #1
011E 7543 0302 BRU *CONWR,RR
011F CC14 0303 TEST2F SSE 4 IF SS#2 IS ON REPEAT TEST
0120 7075 0304 BRU TSTSEL,RR OTHERWISE RETURN TO TEST SEL
0121 7A03 0305 BRU TEST2A

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TEXAS INSTRUMENTS
INCORPORATED
AUSTIN, TEXAS

DOCUMENT NUMBER REVISON
066801-9902 *C

INTIMER - INTERVAL TIMER PCT - 900

SHEET 0009

0122	0000	0306	T2INTA	DATA	0,0	TEST 2 INTERRUPT ROUTINE
0123	0000					
0124	0800	0307	RDSA	RDS	0	READ TIMER
0125	0000	0308		DATA	0	
0126	3800	0309		AND	0000	LOOK AT STATUS ONLY
0127	F000					
0128	0000	0310		CPCL	0000	CORRECT?
0129	0000					
012A	0000	0311		SNE	A	
012B	7A00	0312		BRU	TEST2G	
012C	0000	0313	TEST2L	SSN	0	IF SS#1 IS ON DO NOT PRINT
012D	7A00	0314		BRU	TEST2H	
012E	7541	0315		RRL	*WRITE,RR	PRINT ERROR MESSAGE
P 012F	03B7	0316		DATA	D93	
0130	0F01	0317		LDF	#1	
0131	7543	0318		RRL	*CONVR,RR	
0132	0014	0319	TEST2M	SSR	4	IF SS#2 IS ON REPEAT TEST
0133	7975	0320		BRU	TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
0134	7A00	0321		BRU	TEST2A	
0135	0000	0322	TEST2G	DLN	S+1	SET UP I/O TRAP ADDRESS
0136	0A00	0323		SSB	T2INTR	
P 0137	014A					
0138	A706	0324		DST	#6	
0139	C53H	0325		RMD	M,ST	ENABLE I/O INTERRUPTS
013A	0000	0326		OLDE	0000	
013B	400F					
013C	0020	0327	WDSH	WDS	0	START TIMER/CNT=15
013D	0001	0328		DATA	1	
013E	0000	0329	RDS9	RDS	0	READ TIMER
013F	0000	0330		DATA	0	
0140	3800	0331		AND	0000	LOOK AT STATUS ONLY
0141	0000					
0142	0000	0332		CPCL	0000	CORRECT
0143	1000					
0144	0020	0333		SEN	A	
0145	7A03	0334		BRU	TEST2J	NO
0146	0000	0335		OLDA	0001	
0147	0000					
0148	7544	0336		BRU	*WAIT,BR	WAIT 200MS FOR INTERRUPT
0149	7A00	0337		BRU	TEST2F	NO INTERRUPT
014A	0000	0338	T2INTR	DATA	0,0	
014B	0000					
014C	0000	0339	RDS10	RDS	0	READ TIMER
014D	0000	0340		DATA	0	
014E	3800	0341		AND	0000	LOOK AT STATUS ONLY
014F	F000					
0150	0000	0342		CPCL	0000	CORRECT?
0151	0000					
0152	0020	0343		SEN	A	
0153	7A00	0344		BRU	TEST2I	NO
0154	7A03	0345		BRU	TEST2K	YES
0155	0701	0346	TEST2M	LDA	#1	
0156	7544	0347		RRL	*WAIT,BR	WAIT 1 MILLISECOND

Change 1

TEXAS INSTRUMENTS
INCORPORATED
AUSTIN, TEXAS

DOCUMENT NUMBER REVISION
986801-0902 *C

INTHER - INTERVAL TIMER PDT - 900

SHEET 0010

0157	7802	0348	RRU	TEST2N	
0158	070A	0349	TEST2K	LDA	#10
0159	7544	0350	RRU	*WAIT,RR	WAIT 10 MILLISECONDS
015A	0000	0351	TEST2N	PLDE	=>400F
015B	400F				
015C	0020	0352	WDS7	WDS	0
015D	0001	0353		DATA	1
015E	0000	0354	RDS11	RDS	0
015F	0000	0355		DATA	0
0160	3000	0356		*AND	=>F000
0161	F000				LOOK AT STATUS ONLY
0162	0000	0357		*CPL	=>3000
0163	3000				CORRECT?
0164	0000	0358		SNE	A
0165	7000	0359	BRU	TEST20	YES
0166	0000	0360		*CPL	=>F000
0167	F000				WAIT SOME MORE?
0168	0000	0361		SNE	A
0169	7000	0362	RRU	TEST2M	YES
016A	0000	0363	SSN	0	IF SS#1 IS ON DO NOT PRINT
016B	7004	0364	RRU	TEST2P	
016C	7541	0365	RRU	*WRITE,RR	PRINT ERROR MESSAGE
P 016D	03E5	0366	DATA	D123	
016E	0F01	0367	LDE	#1	
016F	7543	0368	RRU	*CONWR,RR	
0170	CC14	0369	TEST2P	SSR	4
0171	7075	0370	RRU	TSTSEL,RR	IF SS#2 IS ON REPEAT TEST
0172	7002	0371	RRU	TEST2A	OTHERWISE RETURN TO TEST SEL
0173	7541	0372	TEST20	RRU	*WRITE,RR
P 0174	03F9	0373	DATA	D148	PRINT ERROR MESSAGE
0175	CC04	0374	SSN	4	IF SS#2 IS ON REPEAT TEST
0176	7C00	0375	*BRU	TEST2A	
P 0177	00F5				
0178	CC02	0376	SSN	2	IF SS#3 ON RETURN TO TEST SEL
0179	7075	0377	RRU	TSTSEL,RR	

INTHER = INTERVAL TIMER PNT = 999

SHEET 0011

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0378 PEJ
0379 *****
0380 * TEST 3 = LOAD START AND RELOAD TEST *****
0381 *****
0382 TEST3 SSN 8 IF SS#1 IS ON DO NOT PRINT
0383 RRU TEST3A
0384 RRL *WRITE,RR PRINT HEADER
0385 DATA D150
0386 TEST3A LDE #0
0387 WDSR WDS 0 OUTPUT STOP COMMAND
0388 DATA 1
0389 OLD S+1 SFT UP TRAP LOCATION
0390 *SSB T3INT
0391
0392
0393
0394 WDS9 WDS 0 OUTPUT RELOAD/CNT=15
0395 DATA 1
0396 RDS12 RDS 0 READ TIMER
0397 DATA 0
0398 *AND =>F000 LOOK AT STATUS ONLY
0399
0400 *CPL =>1000 CORRECT?
0401
0402
0403
0404 SNE A
0405 RRU TEST3E YES
0406 RMO S,ST DISABLE I/O INTERRUPTS
0407 SSN 8 IF SS#1 IS ON DO NOT PRINT
0408 RRU TEST3D
0409 RRL *WRITE,RR PRINT ERROR MESSAGE
0410 DATA D73
0411 LDE #1
0412 RRL *CONWR,RR
0413 TEST3D LDE #0
0414 WDS10 WDS 0 OUTPUT STOP COMMAND
0415 DATA 1
0416 SSN 4 IF SS#2 ON REPEAT TEST
0417 RRU TEST3A
0418 RRU TSTSEL,RR OTHERWISE RETURN TO TEST SEL
0419 TEST3E *LDA #200
0420
0421
0422 RRL *WAIT,RR WAIT 200MS FOR INTERRUPT
0423 RMO S,ST DISABLE I/O INTERRUPTS
0424 SSN 8 IF SS#1 IS ON DO NOT PRINT
0425 RRU TEST3F
0426 RDS 0 READ TIMER
0427 DATA 0
0428 RRL *WRITE,RR PRINT ERROR MESSAGE
0429 DATA D85
0430 LDE #1
0431 RRL *CONWR,RR

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INTMER = INTERVAL TIMER PDT = 000

SHEET 0012

01A8	0F00	0420	TEST3F	LDE	#0	
01A9	0A20	0427	WDS11	WDS	0	OUTPUT STOP COMMAND
01AA	0001	0428		DATA	1	
01AB	CC94	0429		SSN	4	IF SS#2 ON REPEAT TEST
01AC	7ACE	0430		BRU	TEST3A	
01AD	7075	0431		BRU	TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
01BE	0000	0432	TSINT	DATA	0,0	TEST3 INTERRUPT ROUTINE
01B2	0000					
01B3	0A00	0433	RDS14	RDS	0	READ TIMER
01B4	0000	0434		DATA	0	
01B5	3A00	0435		RAND	=>0000	LOOK AT STATUS ONLY
01B6	0000					
01B7	6000	0436		PCPL	=>0000	CORRECT?
01BA	0000					
01B9	CDAA	0437		SNE	A	
01BA	7A0C	0438		BRU	TEST3H	YES
01BB	CC9A	0439		SSN	8	IF SS#1 IS ON DO NOT PRINT
01BC	7A04	0440		BRU	TEST3I	
01BD	7541	0441		RRL	*WRITE,RR	PRINT ERROR MESSAGE
P 01BE	03B7	0442		DATA	D9S	
01BF	0F01	0443		LDE	#1	
01C0	7543	0444		RRL	*CONWR,RR	
01C1	0F00	0445	TEST3T	LDE	#0	
01C2	0A20	0446	WDS12	WDS	0	OUTPUT STOP COMMAND
01C3	0001	0447		DATA	1	
01C4	CC94	0448		SSN	4	IF SS#2 IS ON REPEAT TEST
01C5	7A0A	0449		BRU	TEST3A	
01C6	7075	0450		BRU	TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
01C7	0F00	0451	TEST3H	LDE	#0	
01C8	0A20	0452	WDS13	WDS	0	OUTPUT STOP COMMAND
01C9	0001	0453		DATA	1	
01CA	7541	0454		RRL	*WRITE,RR	PRINT TEST OK
P 01CB	041C	0455		DATA	D16S	
01CC	CC94	0456		SSN	4	IF SS#2 IS ON REPEAT TEST
01CD	7A0A	0457		BRU	TEST3A	
01CE	CC92	0458		SSN	2	IF SS#3 ON RETURN TO TEST SEL
01CF	7075	0459		BRU	TSTSEL,RR	

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SHEET 0013

```

0460      PEJ
0461      *****
0462      * TEST 4 = SERVICE ERROR TEST
0463      *****
0104      CC98 0464      TEST4 SSN 8          IF SS#1 IS ON DO NOT PRINT
0101      7800 0465      BRU TEST4A
0102      7541 0466      RRL *WRITE,RR      PRINT HEADER
P 0103      0428 0467      DATA D178
0104      0F00 0468      TEST4A LDF #0
0105      0A20 0469      WDS14 WDS 0          OUTPUT STOP COMMAND
0106      0001 0470      DATA 1
0107      8000 0471      OLD 3+1          SET UP I/O TRAP LOCATION
0108      0AC0 0472      *SSB T4INT
P 0109      0201
010A      A706 0473      DST #6
010B      C538 0474      RMD M,ST          ENABLE I/O INTERRUPTS
010C      0A00 0475      *LDE =>400F
010D      400F
010E      0A20 0476      WDS15 WDS 0          START TIMER/CNT=15
010F      0001 0477      DATA 1
010A      0A00 0478      RDS15 RDS 0          READ TIMER
010E      0000 0479      DATA 0
010E      3000 0480      *AND =>F000          LOOK AT STATUS ONLY
010E      F000
010E      6000 0481      *CPL =>1000          CORRECT?
010E      1000
010E      C0A0 0482      SNE A
010E      780A 0483      BRU TEST4C          YES
010E      C548 0484      RMD S,ST          NO = DISABLE I/O INTERRUPTS
010E      CC98 0485      SSN 8          IF SS#1 IS ON DO NOT PRINT
010E      7804 0486      BRU TEST4B
010E      7541 0487      RRL *WRITE,RR      PRINT ERROR MESSAGE
P 010E      0398 0488      DATA D78
010E      0F01 0489      LDF #1
010E      7543 0490      RRL *CONWR,RR
010E      CC94 0491      TEST4B SSN 4          IF SS#2 IS ON REPEAT TEST
010E      78E3 0492      BRU TEST4A
010E      7975 0493      BRU TSTSEL,RR      OTHERWISE RETURN TO TEST SEL
010E      0000 0494      TEST4C *LDA #200
010E      00C8
010E      7544 0495      RRL *WAIT,BR
010E      C548 0496      RMD S,ST          WAIT 200MS FOR I/O INTERRUPTS
010E      CC98 0497      SSN 8          DISABLE INTERRUPTS
010E      7806 0498      BRU TEST4D          IF SS#1 IS ON DO NOT PRINT
010E      0A00 0499      RDS16 RDS 0          READ TIMER
010E      0000 0500      DATA 0
010E      7541 0501      RRL *WRITE,RR      PRINT ERROR MESSAGE
P 010E      03A0 0502      DATA D88
010E      0F01 0503      LDF #1
010E      7543 0504      RRL *CONWR,RR
010E      CC94 0505      TEST4D SSN 4          IF SS#2 IS ON REPEAT TEST
010E      7AD4 0506      BRU TEST4A
010E      7975 0507      BRU TSTSEL,RR      OTHERWISE RETURN TO TEST SEL

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INTHER - INTERVAL TIMER POT - 980

SHEET 0014

0201	0000	0500	T4INT	DATA 0,0	TEST 4 INTERRUPT ROUTINE
0202	0000				
0203	1F00	0500		LDM #0	ZERO COUNT
0204	C533	0510	TEST4F	RIN M,M	INCREMENT COUNT
0205	0000	0511		ALDA #500	
0206	01F4				
0207	7544	0512		BRL #WAIT,0R	WAIT 500MS
0208	C530	0513		RMO M,A	
0209	675A	0514		CPL #00	COUNT > 00?
020A	C0C0	0515		SLE A	
020B	7AFB	0516		BRU TEST4F	YES REPEAT
020C	0000	0517	RDS17	RDS 0	READ TIMER
020D	0000	0518		DATA 0	
020E	1000	0519		ALDM =>110	
020F	0110				
0210	3A00	0520		#AND =>F000	LOOK AT STATUS ONLY
0211	F000				
0212	0000	0521		#CPL =>D000	CORRECT?
0213	D000				
0214	C0A0	0522		SNE A	
0215	7A00	0523		BRU TEST4H	YES
0216	CC00	0524		SSN 0	IF SS#1 IS ON DO NOT PRINT
0217	7804	0525		BRU TEST4G	
0218	7541	0526		BRL #WRITE,0R	PRINT ERROR MESSAGE
P 0219	043C	0527		DATA D100	
021A	0F01	0528		LDE #1	
021B	7543	0529		BRL #CONWR,0R	
021C	CC04	0530	TEST4G	SSN 4	IF SS#2 IS ON REPEAT TEST
021D	7AB6	0531		BRU TEST4A	
021E	7075	0532		BRU TSTSEL,0R	OTHERWISE RETURN TO TEST SEL
021F	C540	0533	TEST4H	RMO S,ST	DISABLE INTERRUPTS
0220	7541	0534		BRL #WRITE,0R	PRINT TEST OK
P 0221	0450	0535		DATA D100	
0222	CC04	0536		SSN 4	IF SS#2 IS ON REPEAT TEST
0223	7800	0537		BRU TEST4A	
0224	CC02	0538		SSN 2	IF SS#3 ON RETURN TO TEST SEL
0225	7075	0539		BRU TSTSEL,0R	

INTMR - INTERVAL TIMER PRT - 990

SHEET 0015

```

0540      PEJ
0541      *****
0542      *      TEST 5 - ILLEGAL FUNCTIONS TEST
0543      *****
0225      CC00 0544      TEST5  SSN  0      IF SS#1 IS ON DO NOT PRINT
0227      7A00 0545      RRU  TEST5A
0228      7541 0546      RRL  *WRITE,RR      PRINT TEST TITLE
P 0229      0460 0547      DATA D200
022A      0F00 0548      TEST5A  LDF  #0
022B      0A20 0549      WDS29  WDS  0      OUTPUT STOP COMMAND
022C      0001 0550      DATA  1
022D      8000 0551      DLD  S+1      SET UP I/O TRAP LOCATION
022E      0A00 0552      @SSB TSINTA
P 022F      024F
0230      A700 0553      DST  #0
0231      C530 0554      RMO  M,ST      ENABLE I/O INTERRUPTS
0232      0A00 0555      @LDE =>4FFF
0233      4FFF
0234      0A20 0556      WDS16  WDS  0      START TIMER/CT=>FFF
0235      0001 0557      DATA  1
0236      0701 0558      LDA  #1
0237      7544 0559      RRL  *WAIT,BR      WAIT 1 MILLISECOND
0238      0A00 0560      @LDE =>8FFF
0239      8FFF
023A      0A20 0561      WDS17  WDS  0      RESTART TIMER/CT=>FFF
023B      0001 0562      DATA  1
023C      0A00 0563      @LDA  #200
023D      0A00
023E      7544 0564      RRL  *WAIT,RR      WAIT 200 MILLISECOND
023F      C540 0565      RMO  S,ST      DISABLE I/O INTERRUPTS
0240      CC00 0566      SSN  0      IF SS#1 IS ON DO NOT PRINT
0241      7A00 0567      RRU  TEST5A
0242      0A00 0568      RDS18  RDS  0      READ TIMER
0243      0000 0569      DATA  0
0244      7541 0570      RRI  *WRITE,RR      PRINT ERROR MESSAGE
P 0245      0475 0571      DATA D210
0246      0F01 0572      LDF  #1
0247      7543 0573      RRL  *CONWR,RR
0248      0F00 0574      TEST5A  LDE  #0
0249      0A20 0575      WDS18  WDS  0      OUTPUT STOP COMMAND
024A      0001 0576      DATA  1
024B      CC04 0577      SSN  4      IF SS#2 IS ON REPEAT TEST
024C      7A00 0578      RRU  TEST5A
024D      7975 0579      RRU  TSTSEL,RR      OTHERWISE RETURN TO TEST SEL
024E      0000 0580      TSINTA  DATA 0,0      INTERRUPT ROUTINE
024F      0000
0250      0A00 0581      RDS19  RDS  0      READ TIMER
0251      0000 0582      DATA  0
0252      3A00 0583      @AND =>F000      LOOK AT STATUS ONLY
0253      F000
0254      0200 0584      @CPL =>0000      CORRECT?
0255      0000
0256      C0A0 0585      SNF  A

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SHEET 0010

0257	7800	0500	RRU	TEST5E	YES
0258	CC00	0507	SSN	8	IF SS#1 IS ON DO NOT PRINT
0259	7804	0508	RRU	TEST5D	
025A	7541	0509	RRL	*WRITE,RR	PRINT ERROR MESSAGE
P 025B	0480	0500	DATA	D223	
025C	0F01	0501	LDE	#1	
025D	7543	0502	RRL	*CONWR,RR	
025E	CC04	0503	TEST5D	SSN	4
025F	7ACA	0504	RRU	TEST5A	IF SS#2 IS ON REPEAT TEST
0260	7075	0505	RRU	TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
0261	0F00	0506	TEST5E	LDE	#0
0262	0A20	0507	WDS19	WDS	0
0263	0001	0508	DATA	1	OUTPUT STOP COMMAND
0264	8000	0509	OLD	S+1	SET UP I/O TRAP LOCATION
0265	0AC0	0600	SSB	T5INTB	
P 0266	0285				
0267	A706	0601	DST	#6	
0268	C53A	0602	RMO	M,ST	ENABLE I/O INTERRUPTS
0269	0A00	0603	OLDE	=>8FFF	
026A	8FFF				
026B	0A20	0604	WDS20	WDS	0
026C	0001	0605	DATA	1	RESTART TIMER/CT=>FFF
026D	0701	0606	LDA	#1	
026E	7544	0607	RRL	*WAIT,BR	WAIT 1 MILLISECOND
026F	0800	0608	OLDE	=>4FFF	
0270	4FFF				
0271	0A20	0609	WDS21	WDS	0
0272	0001	0610	DATA	1	START TIMER/CT=>FFF
0273	0000	0611	OLDA	#200	
0274	00C0				
0275	7544	0612	RRL	*WAIT,BR	WAIT 200MS FOR INTERRUPT
0276	C540	0613	RMO	S,ST	DISABLE I/O INTERRUPTS
0277	CC00	0614	SSN	8	IF SS#1 IS ON DO NOT PRINT
0278	7806	0615	RRU	TEST5F	
0279	0A00	0616	RDS20	RDS	0
027A	0000	0617	DATA	0	READ TIMER
027B	7541	0618	RRL	*WRITE,RR	PRINT ERROR MESSAGE
P 027C	04A3	0619	DATA	D233	
027D	0F01	0620	LDE	#1	
027E	7543	0621	RRL	*CONWR,RR	
027F	0F00	0622	TEST5F	LDE	#0
0280	0820	0623	WDS22	WDS	0
0281	0001	0624	DATA	1	OUTPUT STOP COMMAND
0282	CC04	0625	SSN	4	IF SS#2 IS ON REPEAT TEST
0283	78A6	0626	RRU	TEST5A	OTHERWISE RETURN TO TEST SEL
0284	7075	0627	RRU	TSTSEL,RR	INTERRUPT ROUTINE
0285	0000	0628	T5INTB	DATA	0,0
0286	0000				
0287	0A00	0629	RDS21	RDS	0
0288	0000	0630	DATA	0	READ TIMER
0289	3A00	0631	RAND	=>F000	LOOK AT STATUS ONLY
028A	F000				
028B	8000	0632	OCPL	=>8000	CORRECT?

INTIMER - INTERVAL TIMER PGT - 9A0

SHEET 0017

0280	8000		SNF	A	
0281	CDAD	0633	RRU	TEST5T	YES
0282	7800	0634	SSN	0	IF SS#1 IS ON DO NOT PRINT
0283	CC00	0635	RRU	TEST5H	
0284	7800	0636	BRL	*WRITE,RR	PRINT ERROR MESSAGE
0285	7541	0637	DATA	0241	
P 0286	0400	0638	LDF	#1	
0287	0F01	0639	BRL	*CONWR,RR	
0288	7543	0640	LDF	#0	
0289	0F00	0641	TEST5H		
0290	0820	0642	WDS23	WDS	0
0291	0001	0643	DATA	1	OUTPUT STOP COMMAND
0292	CC04	0644	SSN	4	IF SS#2 IS ON REPEAT TEST
0293	7800	0645	RRU	TEST5A	
0294	7975	0646	RRU	TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
0295	0F00	0647	LDF	#0	
0296	0820	0648	WDS24	WDS	0
0297	0001	0649	DATA	1	OUTPUT STOP COMMAND
0298	8000	0650	OLD	S+1	SET UP TRAP LOCATION
0299	08C0	0651	SSB	TSINTC	
P 0300	028E				
0301	4706	0652	DST	#6	
0302	C53A	0653	RMO	M,ST	ENABLE I/O INTERRUPTS
0303	0800	0654	OLDE	#BFFF	
0304	8FFF				
0305	0820	0655	WDS25	WDS	0
0306	0001	0656	DATA	1	RESTART TIMER/CNT=FFF
0307	0701	0657	LDA	#1	
0308	7544	0658	BRL	*WAIT,RR	WAIT 1 MILLISECOND
0309	0820	0659	WDS26	WDS	0
0310	0001	0660	DATA	1	RESTART TIMER/CNT=FFF
0311	0000	0661	OLDA	#200	
0312	00CA				
0313	7544	0662	BRL	*WAIT,RR	WAIT 200MS FOR INTERRUPT
0314	C54A	0663	RMO	S,ST	DISABLE I/O INTERRUPTS
0315	CC00	0664	SSN	0	IF SS#1 IS ON DO NOT PRINT
0316	7800	0665	RRU	TEST5.I	
0317	0800	0666	RDS	0	READ TIMER
0318	0000	0667	DATA	0	
0319	7541	0668	BRL	*WRITE,RR	PRINT ERROR MESSAGE
P 0320	0400	0669	DATA	0255	
0321	0F01	0670	LDF	#1	
0322	7543	0671	BRL	*CONWR,RR	
0323	0F00	0672	TEST5.I	LDF	#0
0324	0820	0673	WDS27	WDS	0
0325	0001	0674	DATA	1	OUTPUT STOP COMMAND
0326	CC04	0675	SSN	4	IF SS#2 IS ON REPEAT TEST
0327	7C00	0676	BRU	TEST5A	
P 0328	022A				
0329	7975	0677	RRU	TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
0330	0000	0678	TSINTC	DATA	0,0
0331	0000				
0332	0A00	0679	RDS23	RDS	0
					READ TIMER

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INTMER - INTERVAL TIMER PDT - 900

SHEET 0010

02C1	0000	0000	DATA 0	
02C2	3000	0001	0AND 000000	LOOK AT STATUS ONLY
02C3	0000			
02C4	0000	0002	0CPL 000000	CORRECT?
02C5	0000			
02C6	0000	0003	SNE A	
02C7	7000	0004	BRU TEST5M	YES
02C8	0000	0005	SSN 0	IF SS#1 IS ON DO NOT PRINT
02C9	7000	0006	BRU TEST5L	
02CA	7541	0007	BRL *WRITE,RR	PRINT ERROR MESSAGE
P 02CB	0400	0008	DATA D261	
02CC	0001	0009	LDF #1	
02CD	7543	0010	BRL *CONVR,RR	
02CE	0000	0011	LDF #0	
02CF	0020	0012	WDS28 WDS 0	OUTPUT STOP COMMAND
02D0	0001	0013	DATA 1	
02D1	0004	0014	SSN 4	IF SS#2 IS ON REPEAT TEST
02D2	7000	0015	0BRU TEST5A	
P 02D3	022A			
02D4	7075	0016	BRU TSTSEL,RR	OTHERWISE RETURN TO TEST SEL
02D5	7541	0017	BRL *WRITE,RR	PRINT TEST OK
P 02D6	04FF	0018	DATA D271	
02D7	0004	0019	SSN 4	IF SS#2 IS ON REPEAT TEST
02D8	7000	0020	0BRU TEST5A	
P 02D9	022A			
02DA	7075	0021	BRU TSTSEL,RR	OTHERWISE RETURN TO TEST SEL

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INTMR - INTERVAL TIMER PRT - 900

SHEET 0019

```
0700                    PFI
0703                    *****
0704                    *        PRINT STATEMENTS
0705                    *****
0706                    DIS       DATA CRLF, 'INTERVAL TIMER PERFORMANCE DEMONSTRATION'

0204    808A
0205    C9CF
0206    04C5
0207    0206
0208    C1CF
0209    A0D4
0210    C9CF
0211    C5D2
0212    A0D4
0213    C5D2
0214    C6CF
0215    02C0
0216    C1CF
0217    C3C5
0218    A0C4
0219    C5C0
0220    CFCF
0221    03D4
0222    02C1
0223    04C9
0224    CFCF
0225    A0D4    0707                    DATA 'TEST.', CRLF, 0
0226    C5D3
0227    04AF
0228    808A
0229    0200
0230    808A    0708    DIS       DATA CRLF, 'ENTER GROUP NUMBER 0-3 (NORMAL IS 0). '
0231    C5CF
0232    04C5
0233    02A0
0234    C7D2
0235    CFD5
0236    02AC
0237    CFD5
0238    C0C2
0239    C5D2
0240    A0B0
0241    A0B3
0242    A0CF
0243    CFD2
0244    C0C1
0245    CCA0
0246    C0D3
0247    A0B0
0248    A0AF
0249    A0A0    0709                    DATA ' HIT CARRAGE RETURN.', CRLF, 0
0250    C0C0
0251    04A0
0252    C3C1
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INTHER - INTERVAL TIMER PDT - 000

SHEET 0020

0300	0202		
0301	0107		
0302	0300		
0303	0205		
0304	0405		
0305	020F		
0306	FFFF		
0307	808A		
0308	8080		
0309	808A	0710	D28 DATA CRLF, ENTER EXTERNAL REG ADDRESS 00-5F
030A	030F		
030B	0405		
030C	0200		
030D	030F		
030E	0405		
030F	020F		
0310	0100		
0311	0002		
0312	0307		
0313	0001		
0314	0404		
0315	0205		
0316	0303		
0317	0080		
0318	80AD		
0319	8506		
031A	00FF		
031B	00CF	0711	DATA 1 (NORMAL IS 04). HIT CARRAGE RETURN, CRLF, 0
031C	CF02		
031D	0001		
031E	0000		
031F	0903		
0320	0080		
0321	84A9		
0322	AEAD		
0323	00CA		
0324	0904		
0325	00C3		
0326	0102		
0327	0201		
0328	0705		
0329	0002		
032A	0304		
032B	0502		
032C	0EAE		
032D	808A		
032E	0080		
032F	808A	0712	D38 DATA CRLF, CHECK PROGRAM DESCRIPTION FOR
0330	030A		
0331	0503		
0332	0000		
0333	0000		
0334	0000		
0335	0000		
0336	0000		
0337	0000		
0338	0000		
0339	0000		
033A	0000		
033B	0000		
033C	0000		
033D	0000		
033E	0000		
033F	0000		
0340	0000		

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INTMER - INTERVAL TIMER PRT - 900

SHEET 0021

0341	D2C1		
0342	CDA0		
0343	C4C5		
0344	D3C3		
0345	D2C9		
0346	D0D4		
0347	C9CF		
0348	CEA0		
0349	C6CF		
034A	D2FF		
034B	A0D3	0713	DATA 1 SENSE SWITCH SETTINGS.1,CRLF,ENTER TPST 1
034C	C5CF		
034D	D3C5		
034E	A0D3		
034F	D7C9		
0350	D4C3		
0351	CEA0		
0352	D3C5		
0353	D4D4		
0354	C9CE		
0355	C7D3		
0356	AEFF		
0357	B08A		
0358	C5CF		
0359	D4C5		
035A	D2A0		
035B	D4C5		
035C	D3D4		
035D	A0FF		
035E	CE05	0714	DATA 1NUMBER (1-5). HIT CARRAGE RETURN.1,CRLF,0
035F	C0C2		
0360	C5D2		
0361	A0A0		
0362	B1A0		
0363	B5A9		
0364	AEA0		
0365	A0C0		
0366	C9D4		
0367	A0C3		
0368	C1D2		
0369	D2C1		
036A	C7C5		
036B	A0D2		
036C	C5D4		
036D	D5D2		
036E	CEAF		
036F	B08A		
0370	D0D0		
0371	B08A	0715 043	DATA CRLF,1BAD NUMBER, TRY 1-5.1,CRLF,0
0372	C2C1		
0373	C4A0		
0374	CE05		
0375	C0C2		

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INTMER - INTERVAL TIMER PDT = 900

SHEET 0022

0376	C5D2		
0377	ACA0		
0378	D4D2		
0379	09A0		
037A	B1AD		
037B	B5AE		
037C	8D8A		
037D	0000		
037E	8D8A	0716	058 DATA CRLF, 'TEST 1 - START/STOP TEST.', CRLF, 0
037F	D4C5		
0380	D3D4		
0381	APB1		
0382	APAD		
0383	APD3		
0384	D4C1		
0385	02D4		
0386	AFD3		
0387	D4CF		
0388	09A0		
0389	D4C5		
038A	03D4		
038B	AFFF		
038C	8D8A		
038D	0000		
038E	8D8A	0717	068 DATA CRLF, 'STOP ERROR, STATUS IS 1, 0
038F	D3D4		
0390	CFD0		
0391	APC5		
0392	D2D2		
0393	CFD2		
0394	ACD3		
0395	D4C1		
0396	D4D5		
0397	D3A0		
0398	C9D3		
0399	AFFF		
039A	0000		
039B	8D8A	0718	078 DATA CRLF, 'START ERROR, TIMER HAS 1, 0
039C	D3D4		
039D	C1D2		
039E	D4A0		
039F	C5D2		
03A0	D2CF		
03A1	02AC		
03A2	APD4		
03A3	C9CD		
03A4	C5D2		
03A5	APC0		
03A6	C1D3		
03A7	AFFF		
03A8	0000		
03A9	8D8A	0719	088 DATA CRLF, 'NO INTERRUPT, TIMER HAS 1, 0
03AA	CECF		

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INTIMER - INTERVAL TIMER POT - 900

SHEET 0023

03A8	A0C0		
03AC	CE04		
03AD	C502		
03AE	D2D5		
03AF	D0D4		
03B1	ACA0		
03B1	D4C9		
03B2	CDC5		
03B3	D2AD		
03B4	C8C1		
03B5	D3A0		
03B6	D0B0		
03B7	80BA	0720	D05 DATA CRLF, 'READ AFTER INTERRUPT ERROR, TIMER HAS 1,0
03B8	D2C5		
03B9	C1C4		
03BA	A0C1		
03BB	C6D4		
03BC	C5D2		
03BD	A7C9		
03BE	CF04		
03BF	C5D2		
03C1	D2D5		
03C1	D0D4		
03C2	A7C5		
03C3	D2D2		
03C4	CFD2		
03C5	ACA0		
03C6	D4C9		
03C7	CDC5		
03C8	D2AD		
03C9	C8C1		
03CA	D3A0		
03CB	D0B0		
03CC	D3D4	0721	D105 DATA 'START-STOP OK.', CRLF, 0
03CD	C1D2		
03CE	D4AD		
03CF	D3D4		
03D1	CFD2		
03D1	A7CF		
03D2	C8AF		
03D3	80BA		
03D4	D0B0		
03D5	80BA	0722	D118 DATA CRLF, 'TEST 2 - LOAD/START TEST.', CRLF, 0
03D6	D4C5		
03D7	D3D4		
03D8	A7B2		
03D9	A7AD		
03DA	A7C7		
03DB	CFC1		
03DC	C4AF		
03DD	D3D4		
03DE	C1D2		
03DF	D4AD		

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INTIMER - INTERVAL TIMER PNT - 900

SHEET 0024

03E0	D4C5	
03E1	D3D4	
03E2	FFFF	
03E3	8D8A	
03E4	0000	
03E5	8D8A	0723 D128 DATA CRLF, RATE ERROR NOT DETECTED, TIMER HAS 1,0
03E6	D2C1	
03E7	D4C5	
03E8	A0C5	
03E9	D2D2	
03EA	CFD2	
03EB	A0CE	
03EC	CFD4	
03ED	A0C4	
03EE	C5D4	
03EF	C5C3	
03F0	D4C5	
03F1	C4AC	
03F2	A0D4	
03F3	C9CD	
03F4	C5D2	
03F5	A0C8	
03F6	C1D3	
03F7	FFFF	
03F8	0000	
03F9	CCCC	0724 D148 DATA ILOAD AND START TEST OK, 1, CRLF, 0
03FA	C1C4	
03FB	A0C1	
03FC	CFC4	
03FD	A0D3	
03FE	D4C1	
03FF	D2D4	
0400	A0D4	
0401	C5D3	
0402	D4A0	
0403	CFC8	
0404	FFFF	
0405	8D8A	
0406	0000	
0407	8D8A	0725 D158 DATA CRLF, TEST 3 - LOAD START AND RELOAD TEST, 1
0408	D4C5	
0409	D3D4	
040A	A0B3	
040B	A0AD	
040C	A0CC	
040D	CFC1	
040E	C4A0	
040F	D3D4	
0410	C1D2	
0411	D4A0	
0412	C1CE	
0413	C4A0	
0414	D2C5	

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INTMR = INTERVAL TIMER PDT = 900

SHEET 0025

```
0415 C0CF
0416 C1C4
0417 A0D4
0418 C5D3
0419 04AF
041A 8D8A 0726 D15A3 DATA CRLF,0
041B 0000
041C C0CF 0727 D169 DATA 'LOAD START AND RELOAD OK.',CRLF,0
041D C1C4
041E A0D3
041F D4C1
0420 D2D4
0421 A0C1
0422 CFC4
0423 A0D2
0424 C5C0
0425 CFC1
0426 C4A0
0427 CFCB
0428 AFFF
0429 8D8A
042A 0000
042B 8D8A 0728 D179 DATA CRLF,'TEST4 = SERVICE ERROR TEST.',CRLF,0
042C D4C5
042D D3D4
042E B4A0
042F ADA0
0430 D3C5
0431 D2D6
0432 C9C3
0433 C5A0
0434 C5D2
0435 D2CF
0436 D2A0
0437 D4C5
0438 D3D4
0439 AFFF
043A 8D8A
043B 0000
043C 8D8A 0729 D189 DATA CRLF,'BAD SERVICE ERROR STATUS, TIMER HAS 1,0
043D C2C1
043E C4A0
043F D3C5
0440 D2D6
0441 C9C3
0442 C5A0
0443 C5D2
0444 D2CF
0445 D2A0
0446 D3D4
0447 C1D4
0448 D5D3
0449 ACA0
```

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SHEET 0026

044A	D4C0	
044B	CDC5	
044C	D2A0	
044D	CFC1	
044E	D3A0	
044F	0000	
0450	D3C5	0730 D198 DATA 'SERVICE ERROR PROPERLY DETECTED.',CRLF,0
0451	D2D6	
0452	C9C3	
0453	C5A0	
0454	C5D2	
0455	D2CF	
0456	D2A0	
0457	D4D2	
0458	CFD0	
0459	C5D2	
045A	CCD9	
045B	A0C4	
045C	C5D4	
045D	C5C3	
045E	D4C5	
045F	C4AE	
0460	808A	
0461	0000	
0462	808A	0731 D203 DATA CRLF,'TEST 5 = ILLEGAL FUNCTION TEST.',CRLF,0
0463	D4C5	
0464	D3D4	
0465	A0B5	
0466	A0AD	
0467	A0C0	
0468	CCCC	
0469	C5C7	
046A	C1CC	
046B	A0C6	
046C	D5CE	
046D	C3D4	
046E	C9CF	
046F	CFAD	
0470	D4C5	
0471	D3D4	
0472	A0FF	
0473	808A	
0474	0000	
0475	808A	0732 D213 DATA CRLF,'NO ILLEGAL FUNCTION INTERRUPT, TIMER 1
0476	CFCF	
0477	A0C0	
0478	CCCC	
0479	C5C7	
047A	C1CC	
047B	A0C6	
047C	D5CE	
047D	C3D4	
047E	C9CF	

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INTIMER - INTERVAL TIMER PRT - 980

SHEET 0027

047F	CEA0		
0480	C9CF		
0481	D4C0		
0482	D2D2		
0483	D5D0		
0484	D4AC		
0485	AD04		
0486	C9C0		
0487	C5D2		
0488	A0FF		
0489	C8C1	0733	DATA HAS 1,0
048A	D3A0		
048B	0000		
048C	8D8A	0734 0228	DATA CRLF, STATUS AFTER INTERRUPT ERROR, STATUS 1
048D	D3D4		
048E	C1D4		
048F	D5D3		
0490	A0C1		
0491	C6D4		
0492	C5D2		
0493	A0C0		
0494	CFD4		
0495	C5D2		
0496	D2D5		
0497	D0D4		
0498	A0C5		
0499	D2D2		
049A	CFD2		
049B	ACA0		
049C	D3D4		
049D	C1D4		
049E	D5D3		
049F	A0FF		
04A0	C9D3	0735	DATA IS 1,0
04A1	A0FF		
04A2	0000		
04A3	8D8A	0736 0238	DATA CRLF, INO RESTART-START INTERRUPT, TIMER HAS 1,0
04A4	CFCF		
04A5	A0D2		
04A6	C5D3		
04A7	D4C1		
04A8	D2D4		
04A9	AD03		
04AA	D4C1		
04AB	D2D4		
04AC	A0C9		
04AD	CFD4		
04AE	C5D2		
04AF	D2D5		
04B0	D0D4		
04B1	ACA0		
04B2	D4C0		
04B3	CDC5		

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INTMER - INTERVAL TIMER PDT = 980

SHEET 0028

0484	D2AF		
0485	CRC1		
0486	D3AF		
0487	0000		
0488	808A	0737	D249 DATA CRLF, 'ERROR ON RESTART-START INTERRUPT, '
0489	C5D2		
048A	D2CF		
048B	D2AF		
048C	CFCF		
048D	A0D2		
048E	C5D3		
048F	D4C1		
04C0	D2D4		
04C1	A0D3		
04C2	D4C1		
04C3	D2D4		
04C4	A0C9		
04C5	CFD4		
04C6	C5D2		
04C7	D2D5		
04C8	D0D4		
04C9	ACA0		
04CA	D3D4	0738	DATA 'STATUS IS ',0
04CB	C1D4		
04CC	D5D3		
04CD	A0C9		
04CE	D3AF		
04CF	0000		
04D0	8D8A	0739	D258 DATA CRLF, 'NO RESTART-RESTART INTERRUPT, '
04D1	CFCF		
04D2	A0D2		
04D3	C5D3		
04D4	D4C1		
04D5	D2D4		
04D6	A0D2		
04D7	C5D3		
04D8	D4C1		
04D9	D2D4		
04DA	A0C9		
04DB	CFD4		
04DC	C5D2		
04DD	D2D5		
04DE	D0D4		
04DF	ACA0		
04E0	D4C9	0740	DATA 'TIMER HAS ',0
04E1	CDCB		
04E2	D2AF		
04E3	C8C1		
04E4	D3AF		
04E5	0000		
04E6	8D8A	0741	D268 DATA CRLF, 'ERROR ON RESTART-RESTART INTERRUPT, '
04E7	C5D2		
04E8	D2CF		

TEXAS INSTRUMENTS
INCORPORATED
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DOCUMENT NUMBER REVISTON
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INTHER - INTERVAL TIMER PRT - 980

SHEET 0029

04E0	02A0		
04EA	CFCE		
04EB	A0D2		
04EC	C5D3		
04ED	D4C1		
04EE	D2D4		
04EF	A0D2		
04F0	C5D3		
04F1	D4C1		
04F2	D2D4		
04F3	A0C9		
04F4	CFD4		
04F5	C5D2		
04F6	D2D5		
04F7	D0D4		
04F8	ACA0		
04F9	D3D4	0742	DATA 'STATUS IS ' ,0
04FA	C1D4		
04FB	D5D3		
04FC	A0C9		
04FD	D3A0		
04FE	0000		
04FF	C9C0	0743 0278	DATA 'ILLEGAL FUNCTIONS TEST OK' ,CRLF,0
0500	CFCE		
0501	C7C1		
0502	CCA0		
0503	C6D5		
0504	CEC3		
0505	D4C9		
0506	CFCE		
0507	D3A0		
0508	D4C5		
0509	D3D4		
050A	A0CF		
050B	C9AF		
050C	8D8A		
050D	0000		
050E	0000	0744	END START

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INTMER - INTERVAL TIMER PDT - 000

SHEET 0030

A	0000	B	0006	BR	0001	CONWR	0043
CONWRS	0002	CRLF	000A	D00	000A	D10	00F5
D100	00CC	D110	0005	D120	00E5	D140	00F9
D150	0407	D15A0	041A	D160	041C	D170	0420
D180	043C	D190	0450	D20	0315	D200	0462
D210	0475	D220	048C	D230	04A3	D240	0480
D250	0400	D260	04E6	D270	04FF	D30	0330
D40	0371	D50	037E	D60	038F	D70	0390
D80	03A9	D90	0307	E	0001	GROUP	0045
LOC0	0000	LOC0	0030	LOCW	0020	LOOP1	0065
LOOP2	0060	M	0003	RDS1	0005	RDS10	014C
RDS11	015E	RDS12	010A	RDS13	01A5	RDS14	01B3
RDS15	01E0	RDS16	01F0	RDS17	020C	RDS18	0242
RDS19	0250	RDS2	00AF	RDS20	0270	RDS21	0287
RDS22	02B1	RDS23	02C0	RDS3	00C5	RDS4	00D0
RDS5	00E4	RDS6	0101	RDS7	0110	RDS8	0124
RDS9	013E	RECON	0042	RECONS	0001	REGADD	0046
S	0004	ST	0000	START	0000	START0	0000
STARTA	0049	START0	0054	STARTC	0047	T1INT	00CE
T2INTA	0122	T2INT0	014A	T3INT	01B1	T4INT	0201
T3INTA	024E	T3INT0	0205	T3INTC	020E	TEST1	0080
TEST1A	0001	TEST10	00A6	TEST1C	00A3	TEST1D	00BF
TEST1E	00BC	TEST1F	00CB	TEST1G	00E1	TEST1H	00DE
TEST1I	0000	TEST2	00F1	TEST20	0030	TEST2A	00F5
TEST20	0113	TEST2C	0110	TEST2E	0116	TEST2F	011F
TEST2G	0135	TEST2H	0132	TEST2J	0100	TEST2K	0150
TEST2L	012C	TEST2M	0155	TEST2N	010A	TEST20	0173
TEST2P	0170	TEST3	017A	TEST30	003E	TEST3A	017E
TEST30	0100	TEST3E	019F	TEST3F	01A0	TEST3H	01C7
TEST3I	01C1	TEST4	0100	TEST40	003F	TEST4A	01D4
TEST40	01EF	TEST4C	01F2	TEST4D	01FE	TEST4F	0204
TEST4G	021C	TEST4H	021F	TEST5	0226	TEST50	0040
TEST5A	022A	TEST50	0240	TEST5D	025E	TEST5E	0261
TEST5F	027F	TEST5H	0295	TEST5I	0290	TEST5J	0287
TEST5L	02CE	TEST5M	02D5	TST5EL	0075	WAIT	0044
WAIT0	0003	WDS1	0003	WDS10	010A	WDS11	01AC
WDS12	01C2	WDS13	01C0	WDS14	01D5	WDS15	01DE
WDS16	0234	WDS17	023A	WDS18	0249	WDS19	0262
WDS2	00AD	WDS20	0260	WDS21	0271	WDS22	0280
WDS23	0296	WDS24	029C	WDS25	02A5	WDS26	02A9
WDS27	02B0	WDS28	02CF	WDS29	0220	WDS3	00E2
WDS4	00F6	WDS5	00FF	WDS6	013C	WDS7	015C
WDS8	017F	WDS9	0100	WRITE	0041	WRITES	0000
X	0002						

0000 ERRORS